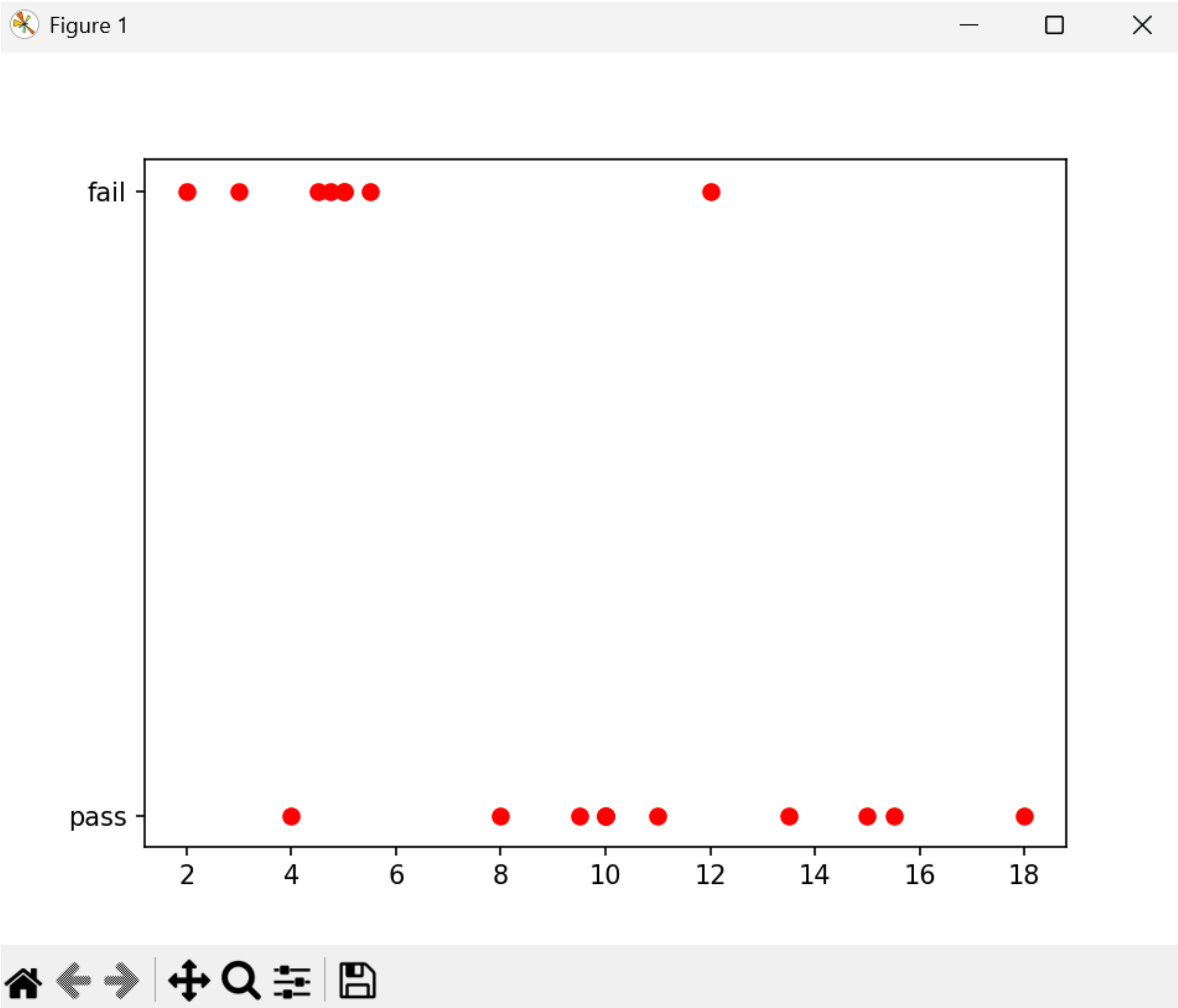
Machine Learning-Lecture 6

P1.py

```
C:\ml\L6>python p1.py
(100, 4)
  location   area  bedrooms      price
0     thane  527.0       2.0   85.700449
1    bandra  782.0       NaN  419.200371
2  borivali  977.0       1.0  190.360680
3  borivali  578.0       3.0  254.243379
4     dadar  791.0       3.0  244.845905
  location   area  bedrooms      price
95   bandra  824.0       3.0  436.795968
96   bandra  733.0       NaN  230.471941
97    dadar    NaN       1.0   97.154017
98   bandra  908.0       1.0  254.692142
99   bandra  664.0       3.0  387.506794
location    0
area        5
bedrooms    5
price       0
dtype: int64
0.1282908114563437
0.12030875237470681
location    0
area        0
bedrooms    0
price       0
dtype: int64
   location        area  bedrooms
0     thane  527.000000  2.000000
1    bandra  782.000000  1.936842
```

```
2   borivali  977.000000  1.000000
3   borivali  578.000000  3.000000
4      dadar  791.000000  3.000000
..       ...         ...       ...
95    bandra  824.000000  3.000000
96    bandra  733.000000  1.936842
97     dadar  747.063158  1.000000
98    bandra  908.000000  1.000000
99    bandra  664.000000  3.000000
[100 rows x 3 columns]
          area  bedrooms  ...  location_dadar  location_thane
0   527.000000  2.000000  ...           False            True
1   782.000000  1.936842  ...           False           False
2   977.000000  1.000000  ...           False           False
3   578.000000  3.000000  ...           False           False
4   791.000000  3.000000  ...            True           False
..         ...       ...  ...             ...             ...
95  824.000000  3.000000  ...           False           False
96  733.000000  1.936842  ...           False           False
97  747.063158  1.000000  ...            True           False
98  908.000000  1.000000  ...           False           False
99  664.000000  3.000000  ...           False           False
[100 rows x 6 columns]
0.931090146160168
enter area : 800
enter number of  bedrooms: 2
 1.bandra ,2.borivali,3. dadar,4. thane 3
[187.31557629]
```

P2.py



P3.py

C:\ml\L6>python p3.py

```
   no  hours result
0   1  18.00   pass
1   2   3.00   fail
2   3   5.00   fail
3   4  10.00   pass
4   5  10.00   pass
5   6   5.00   fail
6   7   4.75   fail
7   8  10.00   pass
8   9  13.50   pass
```

9   10   2.00   fail

10 11   4.00   pass

11 12 15.00   pass

12 13 12.00   fail

13 14 15.50   pass

14 15 11.00   pass

15 16   4.50   fail

16 17   5.50   fail

17 18   8.00   pass

18 19   9.50   pass

enter hours 8

['pass']

P4.py

C:\ml\L6>python p4.py

    no hours result

0   1 18.00   pass

1   2   3.00   fail

2   3   5.00   fail

3   4 10.00   pass

4   5 10.00   pass

5   6   5.00   fail

6   7   4.75   fail

7   8 10.00   pass

8   9 13.50   pass

9   10   2.00   fail

10 11   4.00   pass

11 12 15.00   pass

12 13 12.00   fail

13 14 15.50   pass

14 15 11.00   pass

15 16 4.50 fail

16 17 5.50 fail

17 18 8.00 pass

18 19 9.50 pass

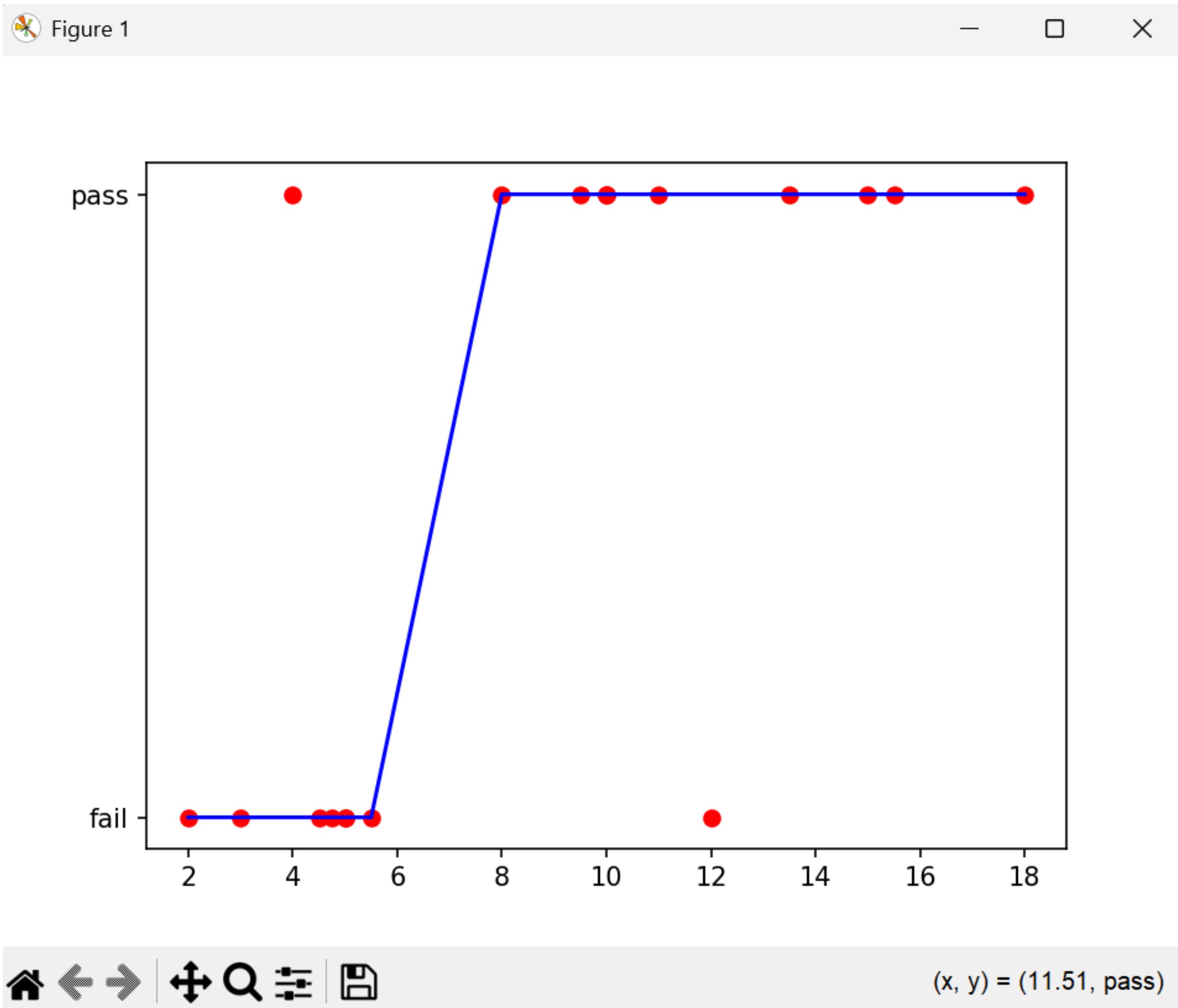
['fail' 'pass']

enter hours 7

[0.4776184]

Fail

P5.py



P6.py

```
C:\ml\L6>python p6.py
[[ 2.]
 [12.]
 [10.]
 [ 8.]]
9     fail
12    fail
3     pass
17    pass
Name: result, dtype: object
['fail' 'pass' 'pass' 'pass']
[[1 1]
 [0 2]]
              precision    recall  f1-score   support

        fail       1.00      0.50      0.67         2
        pass       0.67      1.00      0.80         2

    accuracy                           0.75         4
   macro avg       0.83      0.75      0.73         4
weighted avg       0.83      0.75      0.73         4
```